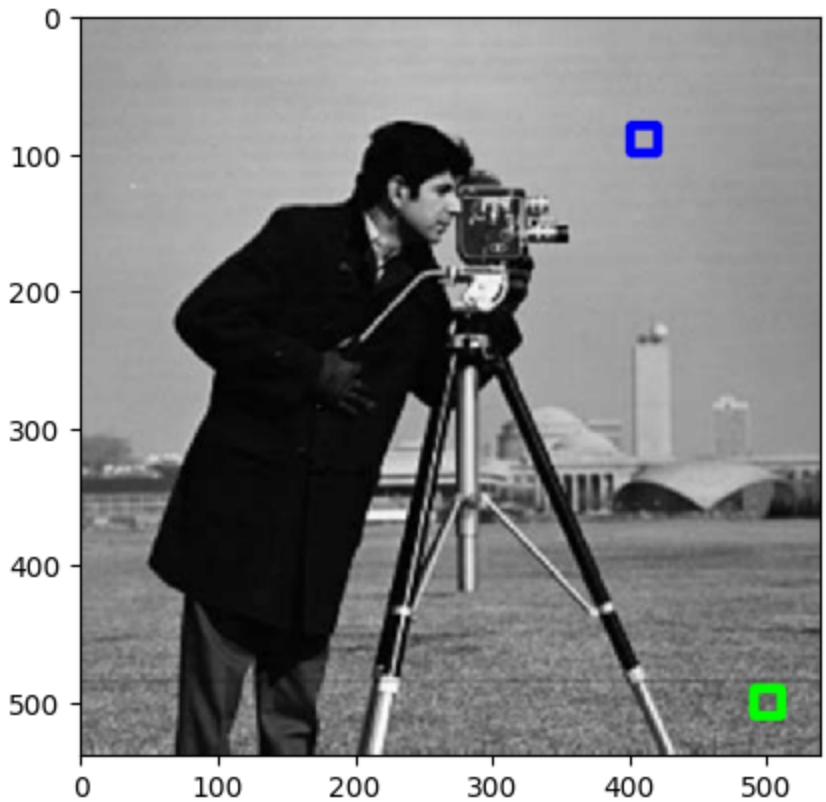
INSTITUTO SUPERIOR TECNOLÓGICO QUITO

Excelencia en Educación Superior

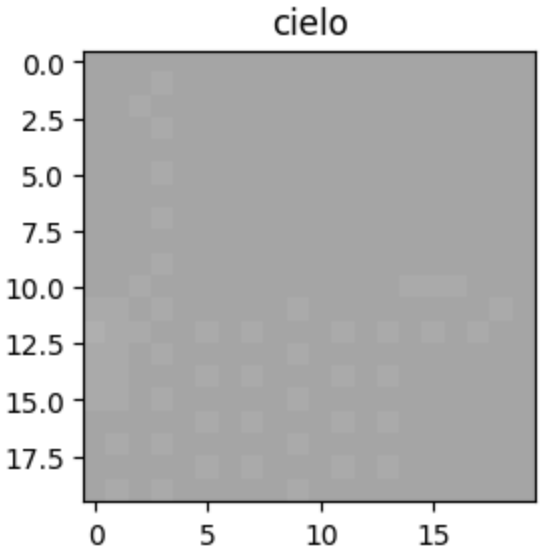
4.1 Descriptores de Textura

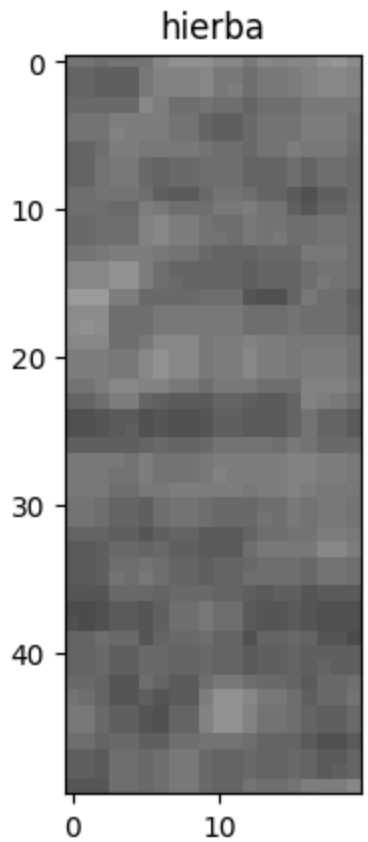
Erick Mejia

In [20]:
```
# En primer lugar, obtenemos la máscara de la imagen original mediante la umbraliza
import matplotlib.pyplot as plt
import cv2
import numpy as np
img = cv2.imread('images/photographer.png')

window = cv2.rectangle(img.copy(), (400,80), (420,100), (0, 0, 255), 5)
window = cv2.rectangle(window, (490,490), (510,510), (0, 255, 0), 5)
plt.imshow(window, cmap='gray')
plt.show()
```



In [21]:
```
# Seleccionamos una región del cielo y una de la hierba
cielo = img[80:100, 400:420]
hierba = img[460:510, 490:510]

fig, ax = plt.subplots(1,2)
ax[0].imshow(cielo, cmap='gray')
ax[0].set_title('cielo')
ax[1].imshow(hierba, cmap='gray')
ax[1].set_title('hierba')
plt.show()
```





MATRIZ DE COOCURRENCIA

In [22]:
```
# MATRIZ DE COOCURRENCIA (Gray-Level Cooccurence Matrix - GLCM)
from tabulate import tabulate
from skimage.feature import graycomatrix, graycoprops
headers = ['','cielo', 'hierba']
CN, DIS, H, ASM, E, COR = ['contraste'], ['disimilitud'], ['homogeneidad'], ['ASM']

patches = []
patches.append(cielo)
patches.append(hierba)

for i in range(len(patches)):
    image = cv2.cvtColor(patches[i], cv2.COLOR_BGR2GRAY)
    GLCM = graycomatrix(image, distances=[5], angles=[0], levels=256, symmetric=Tru
    CN.append(graycoprops(GLCM, 'contrast')[0,0])
    DIS.append(graycoprops(GLCM, 'dissimilarity')[0,0])
    H.append(graycoprops(GLCM, 'homogeneity')[0,0])
    ASM.append(graycoprops(GLCM, 'ASM')[0,0])
    E.append(graycoprops(GLCM, 'energy')[0,0])
    COR.append(graycoprops(GLCM, 'correlation')[0,0])

my_data = [tuple(CN), tuple(DIS), tuple(H), tuple(ASM), tuple(E), tuple(COR)]

print(tabulate(my_data, headers=headers))
```

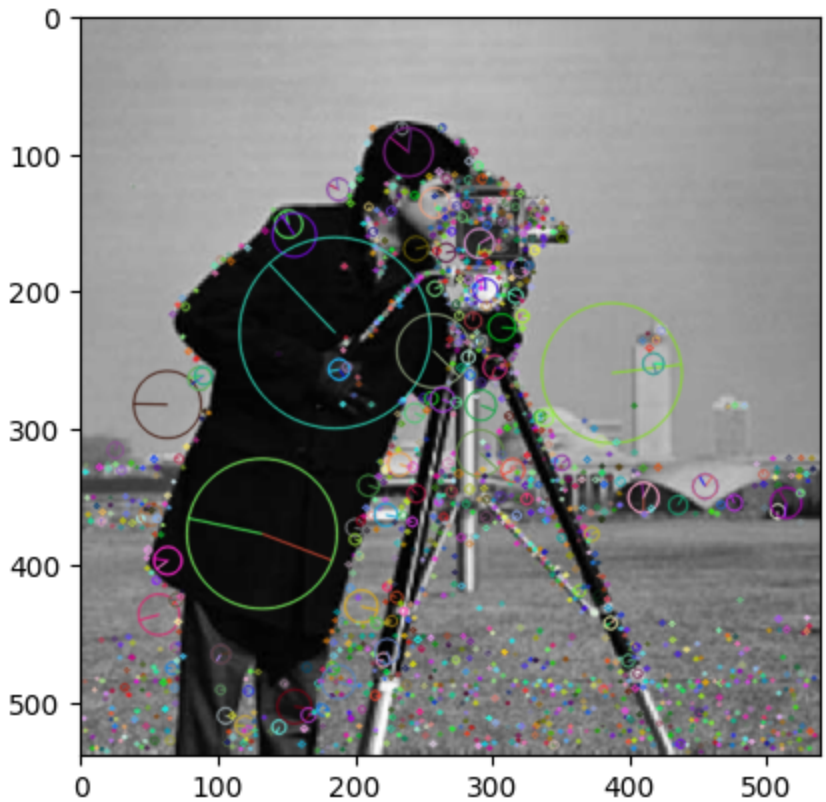
```
                 cielo       hierba
------------  --------  -----------
contraste     3.68      274.288
disimilitud   1.4        12.616
homogeneidad  0.512471    0.095545
ASM           0.142717    0.002824
Energía       0.377779    0.0531413
Correlación   0.190521    0.339645
```

Scale-Invariant Feature Transform

In [28]:

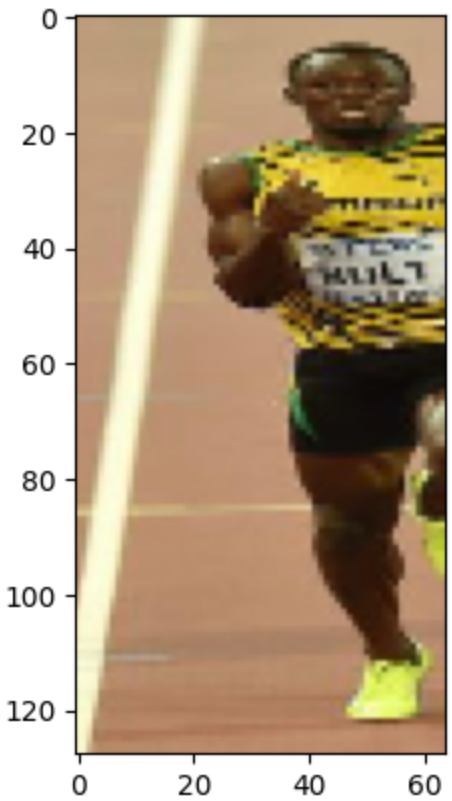
```
# SIFT algorithm
img = cv2.imread('images/photographer.png')

# Definición del algoritmo
sift = cv2.SIFT_create()

# Identificamos los puntos de interés
kp = sift.detect(img, None)

# Extraemos el descriptor para obtener el fingreprint
kp, des = sift.compute(img, kp)
print(np.shape(des))
# print(des) # Matriz donde las filas son los keypoints y las columnas las caracter

# Visualizamos los puntos de interés
img=cv2.drawKeypoints(img, kp, img, flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINT
plt.imshow(img)
plt.show()
```

```
(1672, 128)
```

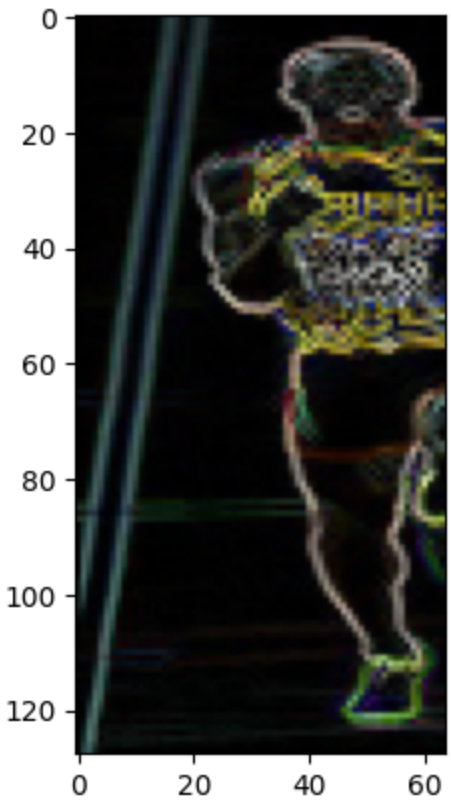


Histogram of Oriented Gradients

In [29]:
```
# HISTOGRAM OF ORIENTED GRADIENTS

# En primer lugar, obtenemos la máscara de la imagen original mediante la umbraliza
from skimage import io
img = io.imread('images/pedestrian.jpg')
img = np.float32(img/255)
crop = img[75:550,475:600]
crop = cv2.resize(crop, (64, 128))

plt.imshow(crop, cmap='gray')
plt.show()
```



In [30]:
```
# Calculamos el gradiente con el kernel de Sobel
crop = np.float32(crop)
gx = cv2.Sobel(crop, cv2.CV_32F, 1, 0, ksize=1)
gy = cv2.Sobel(crop, cv2.CV_32F, 0, 1, ksize=1)

# Calculamos la magnitud y la dirección del gradiente
mag, angle = cv2.cartToPolar(gx, gy, angleInDegrees=True)

# Visualizamos la imagen obtenida en la magnitud del gradiente
mag = mag/np.max(mag)
plt.imshow(mag, cmap='gray')
plt.show()
```



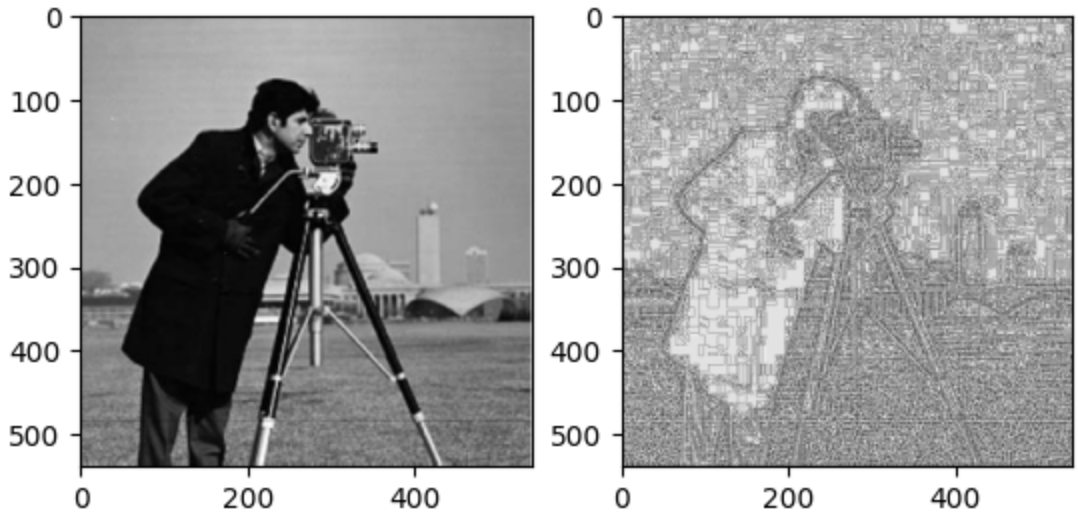
In [31]:
```
# Cálculo del descriptor: Histogram of Oriented Gradients
from skimage.feature import hog
H = hog(crop, orientations=9, pixels_per_cell=(8, 8), cells_per_block=(2, 2), trans
np.shape(H)
```

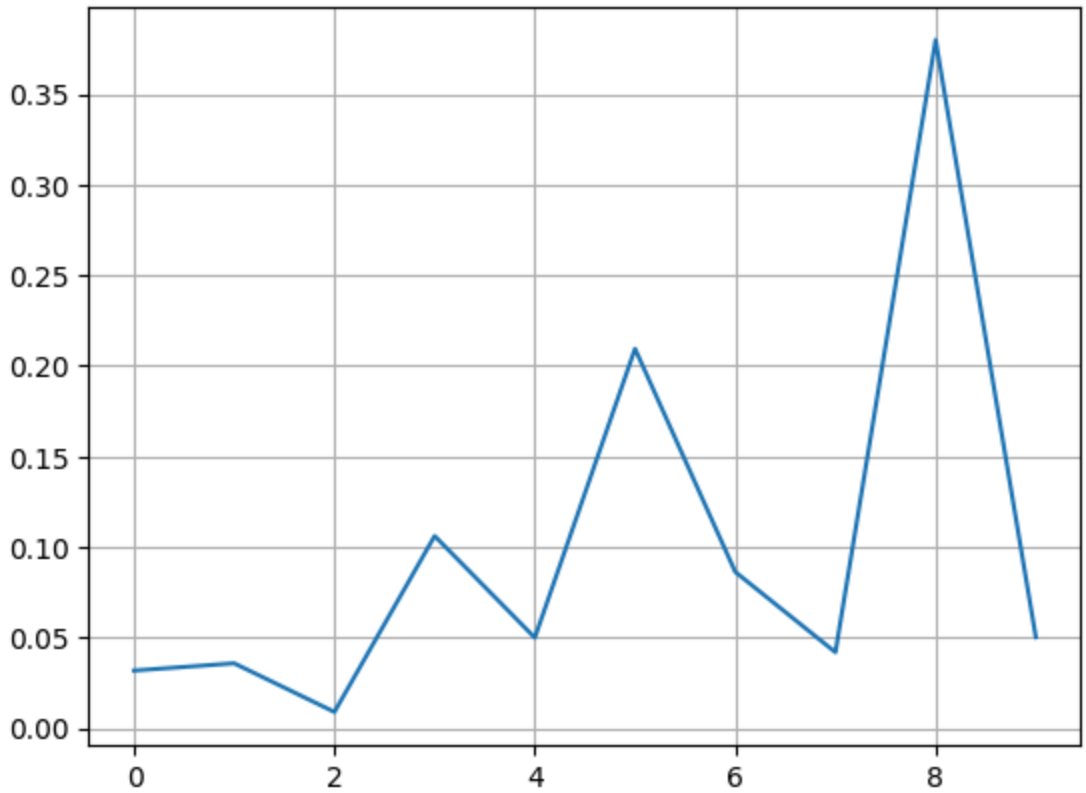
Out[31]: (3780,)

Local Binary Patterns

In [33]:
```
# LOCAL BINARY PATTERNS (LBP)
from skimage.feature import local_binary_pattern
img = cv2.imread('images/photographer.png')

# Definimos los parámetros iniciales
R = 1 # Radio
P = 8*R # Número de píxeles vecinos
epsilon = 1e-7 # Para normalizar el histograma

# Extraemos la imagen LBP
img_gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
lbp_image = local_binary_pattern(img_gray, P, R, method='uniform')
a,b = np.unique(lbp_image, return_counts=True)

# Calculamos el histograma
lbp_image = np.uint8(lbp_image)
hist = cv2.calcHist([lbp_image.ravel()], [0], None, [P+2], [0, P+2])

# Normalizamos el histograma
hist = hist.astype("float")
```

```
hist /= (hist.sum() + epsilon)

# Los P+2=10 bins del histograma serán los 10 atributos que nos permitirán caracter

# Visualizamos la imagen LBP
figs, ax = plt.subplots(1,2)
ax[0].imshow(img, cmap='gray')
ax[1].imshow(lbp_image, cmap='gray')
plt.show()

# Visualizamos el histograma
plt.plot(hist)
plt.grid(True)
plt.show()
```





5. Github

https://github.com/Erick-305/machine-learning.git